Cycle Test 1: worked solutions

29 Aug 2024 · 20 marks · 1 hour · CEPC12

Q1

2 marks

> Explain the concepts of vapour pressure and cavitation.

- **Vapour pressure**: when a liquid is in a closed space, molecules escape from its surface as vapour; at equilibrium, the **partial pressure exerted by this vapour on the liquid surface** is the vapour pressure. It **increases with temperature** (water: 2.34 kPa at 20 °C; 101.3 kPa at 100 °C). A liquid boils when the surrounding pressure falls to its vapour pressure.
- **Cavitation**: when the pressure at some point in a flowing liquid **falls below its vapour pressure**, the liquid boils locally and **vapour bubbles (cavities)** form. Carried to a higher-pressure region, the bubbles **collapse suddenly**, producing very high local pressures. Results: **pitting and erosion** of metal surfaces, noise, vibration and loss of efficiency. Occurs in pump suctions, turbine runners and draft tubes, propellers, spillways. Avoided by keeping pressures above vapour pressure (limit suction head, use a cavitation-resistant material).

Q2

4 marks

> What is the effect of temperature on the viscosity of water and that of air?
>
> A vertical gap 2.2 cm wide of infinite extent contains a fluid of viscosity 2.0 N·s/m^2 and S = 0.9. A metallic plate 1.2 m × 1.2 m × 0.2 cm is to be lifted up at a constant velocity of 0.15 m/s through the gap. If the plate is in the middle of the gap, find the force required. Weight of the plate = 40 N.

Temperature effect:

- **Liquids (water)**: viscosity **decreases** as temperature rises. Viscosity in liquids comes mainly from **cohesion** between molecules, which weakens on heating.

- **Gases (air)**: viscosity **increases** as temperature rises. Viscosity in gases comes from **molecular momentum exchange**, which increases as molecules move faster.

Numerical. The plate (0.2 cm thick) sits in the middle of the 2.2 cm gap, so each side has a gap of $(2.2 - 0.2)/2 =$ **1.0 cm**.

Shear force on one face = $\mu A V / y = 2.0 \times (1.2 \times 1.2) \times 0.15 / 0.01 = 43.2$ N

Total viscous drag (both faces) = $2 \times 43.2 =$ **86.4 N** (acts downward, opposing motion)

Upthrust = $\rho g \times$ volume $= 0.9 \times 9810 \times (1.2 \times 1.2 \times 0.002) =$ **25.43 N** (upward)

Force required F = drag + weight − upthrust = 86.4 + 40 − 25.43 =

100.97 N ≈ 101 N

Q3

4 marks

> Explain the hydrostatic law. The diameters of the small and large pistons of a hydraulic jack are 2 cm and 10 cm. A force of 60 N is applied on the small piston. Find the load lifted by the large piston when (a) the pistons are at the same level, (b) the small piston is 20 cm above the large piston. Density of liquid = 1000 kg/m^3.

Hydrostatic law: in a fluid at rest, the rate of increase of pressure in the vertically downward direction equals the specific weight of the fluid: $\partial p/\partial z = -\rho g$ (z upward). Integrating, $p = \rho g h$ at depth h below the free surface. So pressure depends only on depth, not on the shape of the container.

$A_{small} = \pi/4 \times 0.02^2 = 3.1416 \times 10^{-4}$ m^2; $A_{large} = \pi/4 \times 0.10^2 = 7.854 \times 10^{-3}$ m^2.

Pressure under the small piston $p = 60 / 3.1416 \times 10^{-4} = 190{,}986$ N/m^2

(a) Same level: $W = p \times A_{large} = 190{,}986 \times 7.854 \times 10^{-3} =$ **1500 N**

(b) Small piston 20 cm higher: pressure at the large piston = $p + \rho g h = 190{,}986 + 1000 \times 9.81 \times 0.2 = 192{,}948$ N/m^2

$W = 192{,}948 \times 7.854 \times 10^{-3} =$ **1515.4 N**

Q4

5 marks

> Derive an expression for the total pressure and the position of the centre of pressure on a curved surface submerged in a liquid.
>
> A circular plate 3.0 m in diameter has a concentric circular hole 1.5 m in diameter and is immersed in water so that its greatest and least depths below the free surface are 4 m and 1.5 m. Find (a) the total pressure on one face and (b) the position of the centre of pressure.

Curved surface. On a small area dA at depth h, the force dF = ρgh dA acts normal to the surface, so the direction changes from point to point. Resolve each dF into horizontal and vertical components and add them separately.

- **Horizontal component**: $F_x = \int \rho g h \, dA \cos\theta = \int \rho g h \, dA_x$, where $dA_x = dA\cos\theta$ is the projection of dA on a vertical plane. So **F_x = total pressure on the vertical projection of the curved surface** = $\rho g A_{proj} \bar{h}_{proj}$, acting through the **centre of pressure of that projected area** ($h^* = \bar{h} + I_G/(A\bar{h})$).
- **Vertical component**: $F_y = \int \rho g h \, dA \sin\theta = \rho g \int h \, dA_y$ = ρg × (volume of liquid between the curved surface and the free surface). So **F_y = weight of liquid vertically above the curved surface** (real or imaginary), acting through the **centroid of that volume**.
- **Resultant**: $\boxed{F = \sqrt{(F_x^2 + F_y^2)}}$, inclined at **$\tan\theta = F_y/F_x$** to the horizontal. For a circular surface it passes through the centre of curvature.

Numerical. Annular plate: D = 3 m, d = 1.5 m, inclined so that the depth changes by 4 − 1.5 = 2.5 m over the 3 m diameter: $\sin\theta = 2.5/3 = 0.8333$.

$A = \pi/4\,(3^2 - 1.5^2) = 5.301\ m^2$; centroid depth $\bar{h} = (4 + 1.5)/2 =$ **2.75 m**

(a) $F = \rho g A\bar{h} = 9810 \times 5.301 \times 2.75 =$ $\boxed{\textbf{143.02 kN}}$

$I_G = \pi/64\,(3^4 - 1.5^4) = \pi/64\,(81 - 5.0625) = 3.728\ m^4$

(b) $h^* = \bar{h} + I_G \sin^2\theta/(A\bar{h}) = 2.75 + 3.728 \times 0.6944/(5.301 \times 2.75) = 2.75 + 0.178$

$h^* =$ $\boxed{\textbf{2.928 m below the free surface}}$

Q5

5 marks

> What do you mean by single column manometers? Determine the pressure difference ($p_M - p_N$) for the manometer shown (oil of RD 0.8 above the mercury).

Single column manometer: a modified U-tube manometer in which **one limb is replaced by a reservoir of large cross-sectional area** (about 100 times the tube area). When pressure is applied, the fall of level in the reservoir is very small and can be neglected (or corrected for), so the pressure is read from the **rise of liquid in the narrow limb only**. Types: **vertical** and **inclined** (more sensitive, for small pressures). For a vertical one: $p_A = h_2\rho_2 g - h_1\rho_1 g$ (with a reservoir correction $\delta h(\rho_2 - \rho_1)g$ if needed).

Compound manometer. All tubing above the mercury is filled with oil ($\rho_o = 800$ kg/m^3). Reading the figure, measured from X–X (the mercury surface in the limb connected to M):

- M is 20 + 16 = **36 cm** above X–X
- mercury in the second limb stands **16 cm** above X–X
- mercury in the third limb is 12 cm lower, i.e. **4 cm** above X–X
- mercury in the fourth limb is 15 cm higher than the third, i.e. **19 cm** above X–X
- N is 15 cm above that, i.e. **34 cm** above X–X

Moving from M to N (down adds ρgh, up subtracts):

$$p_M + \rho_o g(0.36) - \rho_{Hg} g(0.16) + \rho_o g(0.12) - \rho_{Hg} g(0.15) - \rho_o g(0.15) = p_N$$

$$p_M - p_N = \rho_{Hg} g(0.16 + 0.15) - \rho_o g(0.36 + 0.12 - 0.15)$$

$$= 9810 \times (13.6 \times 0.31 - 0.8 \times 0.33) = 9810 \times (4.216 - 0.264)$$

$p_M - p_N =$ **$38{,}769$ N/m^2 ≈ 38.8 kPa** (M is at the higher pressure)